FAE MONSOON 2026 | PROJECT STAGE 1

Spatial Audio Conferencing

Every voice in a group call, heard from its own seat at the table



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THE PROBLEM

In a group call, every voice comes from the same place

1

Hard to tell who is speaking

When people talk together, voices blend into one stream.

2

Listening takes more effort

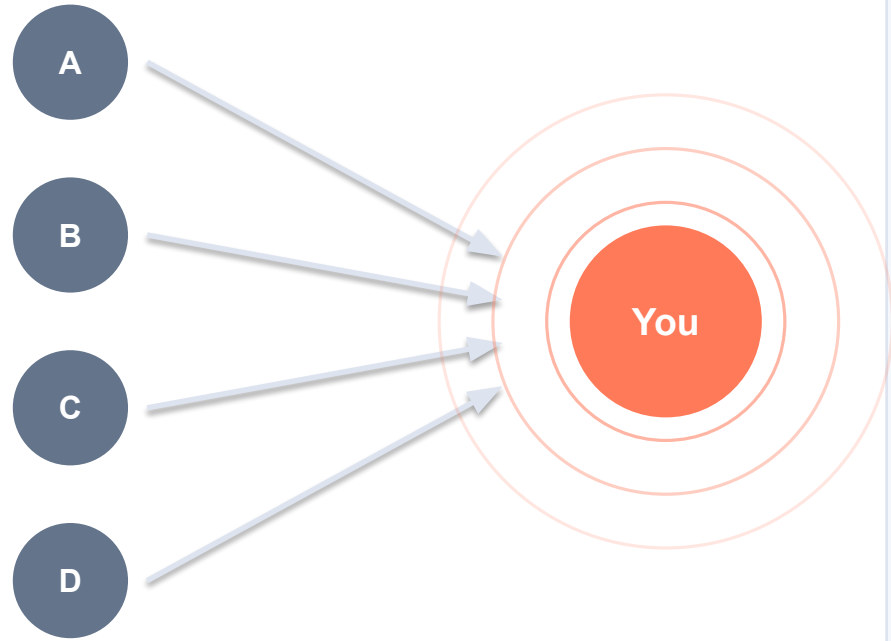
Following overlapping speakers is tiring over a long call.

3

Not how we naturally hear

In a room we separate voices by direction (the "cocktail party effect"). A call removes it.

A normal call: one mono stream



All voices arrive at one point

02

OUR IDEA

A virtual round table: pick a seat, hear from there

Pick a seat

Choose a position on a table or room layout; everyone sees the layout.

Hear it from there

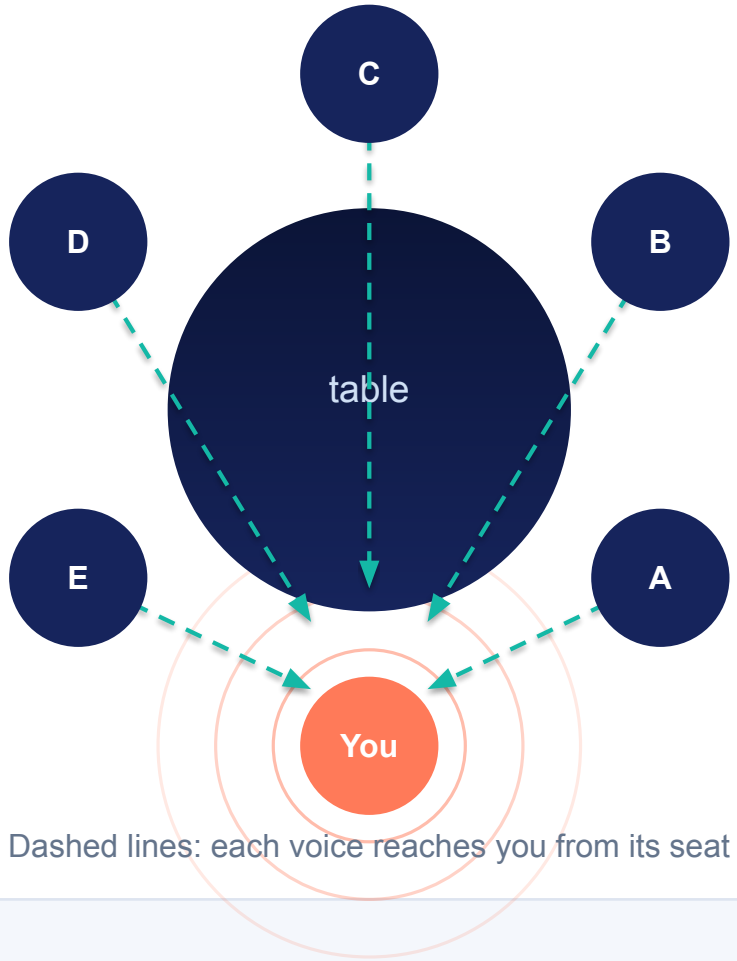
Each voice is processed to appear from that person's seat.

Your own mix

Turn any person up or down with a volume control.

Android or Windows app

Runs on a phone or laptop with headphones.



Dashed lines: each voice reaches you from its seat

03

HOW IT WORKS

From microphone to a spatial mix in your headphones

1. **Microphone** — each user speaks
2. **Send** — one stream per user (WebRTC, Opus)
3. **Receive** — every voice kept separate
4. **Spatial engine** — seat angle sets delay, level and filter
5. **Mix** — add voices, volume per person
6. **Headphones** — left and right ear signals

FAE concepts we apply

Digital audio

48 kHz mono voice streams

Level and dB

gain per person, headroom

Panning and delay

ear-to-ear level and sub-millisecond delay

Filtering (HRTF)

direction changes the spectrum

Mixing

sum voices, limiter on the output

Latency

aim under 150 ms one way (ITU-T G.114)

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OPTIONAL SECOND PHASE

Focus mode: the sound follows where you look

1

Face tracking

the camera estimates which seat you face (open-source face model).

2

Voice thrown

your voice is louder for the person you face.

3

Voice received

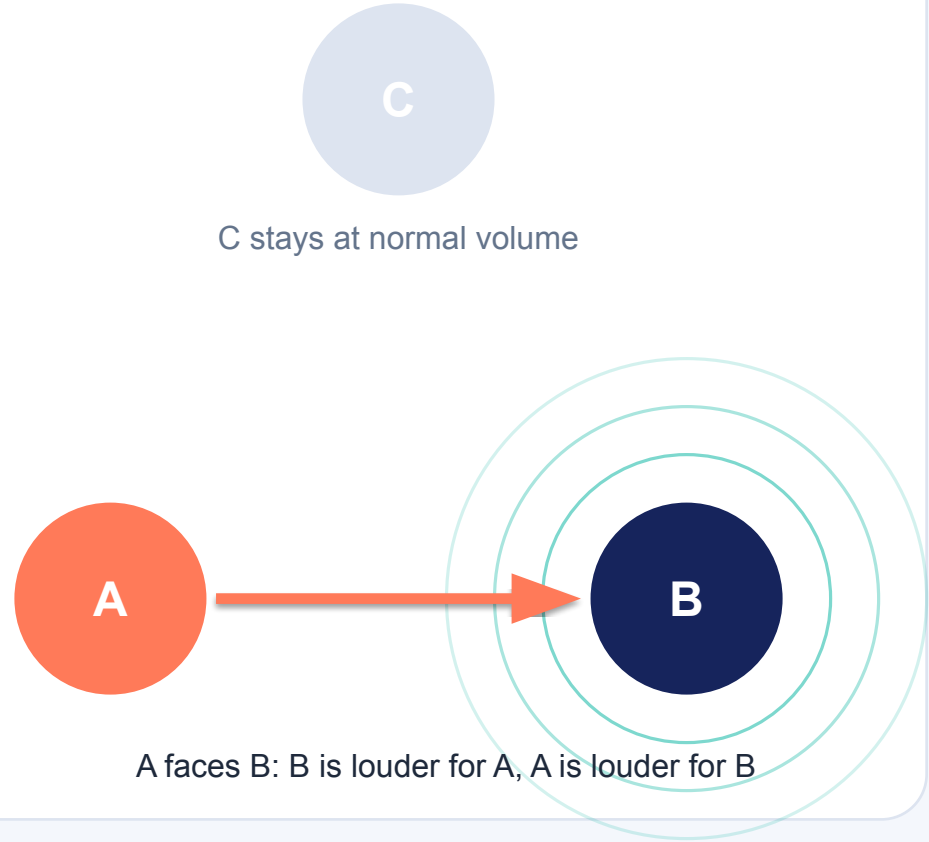
the person you look at sounds louder to you.

4

Safe fallback

camera off or unreliable? Set focus by hand.

One-to-one focus inside a group call



05

SOCIAL RELEVANCE AND FEASIBILITY

Useful for many people, built with free tools

Who benefits

Classrooms and study groups

follow who is speaking in online and hybrid discussion

Meetings of 3 to 30 people

board meetings, business meetings, project teams

Inclusion

people who find overlapping voices hard to follow may benefit (to be tested)

Open-source toolkit

Calls	**WebRTC**
Voice codec	**Opus**
Spatial audio	**Web Audio API or OpenAL Soft**
Face direction	**MediaPipe**
Prototype and analysis	**Python, Audacity**

No extra hardware: just a phone or laptop and headphones

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PLAN AND TESTING

From concept to a tested demo

Stage 1
Concept and scope

- Define the idea
- Choose open-source tools
- Plan the tests

Stage 2
Prototype

- Seat layout screen
- Spatial mixer, volume per person
- Test with recorded voices

Stage 3
Working demo

- Live call, several users
- Face-tracking focus mode
- Listening test with volunteers

How we will check it

- Listening test: spatial audio vs a normal call
- Can listeners tell who is speaking?
- Do they follow overlapping speech better?
- Delay under 150 ms, no clipping

Main risks: delay, headphones needed for the full effect, face-tracking accuracy (manual control as backup).

Thank you